

Lab 8 - The Law of Reflection

Learning Objectives:

1. *To understand how we see objects*
2. *To understand the law of reflection*
3. *To understand the image formation in plane mirror*

I. Observation Experiment: Law of Reflection

Most objects reflect a certain portion of the light falling on them. The goal of the following experiment is to study the law of reflection.

Equipment:

- Optics Bench with Light Source
- Ray Table and Base
- Component Holder
- Slit Plate
- Slit Mask
- Ray Optics Mirror

Introduction

The shape and location of the image created by reflection from a mirror of any shape is determined by just a few simple principles. One of these principles you already know: light propagates in a straight line. In this experiment, you will observe the reflection of a single ray of light from a plane mirror.

To get started, please answer the following questions.

1. Describe why you cannot see an object in a mirror when the lights are off.

The image shown in a mirror is the result of light reflecting off an object, into the mirror's surface and into your eye

2. Why can you not see the object when something is blocking it even though the lights are on?

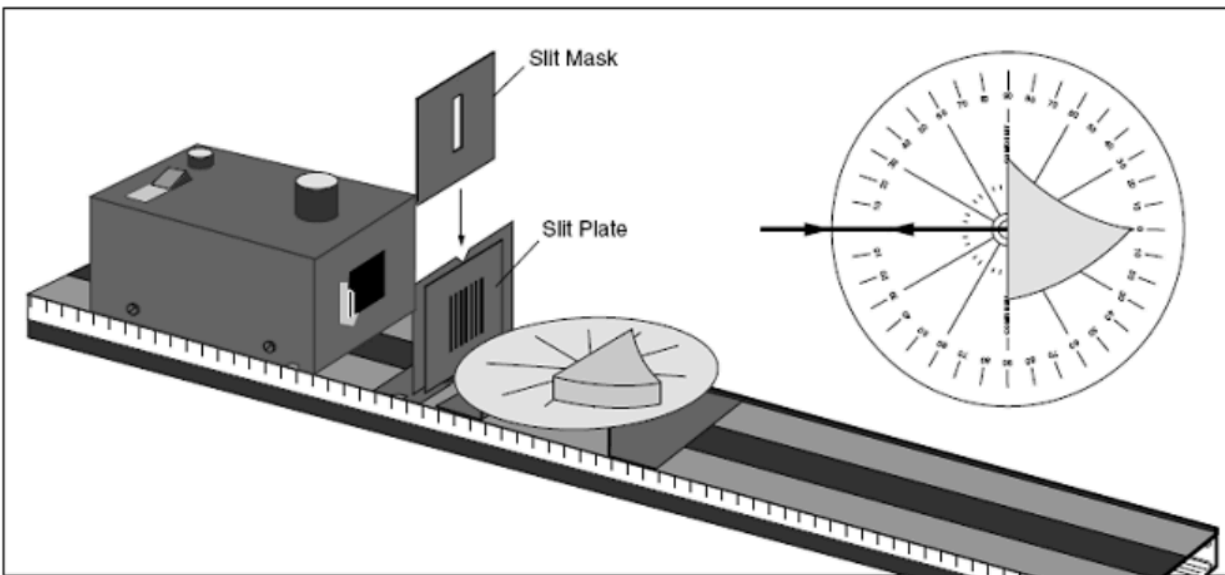
The path of light reflected off the object is obscured on its trajectory from the object to the mirror.

Part 1

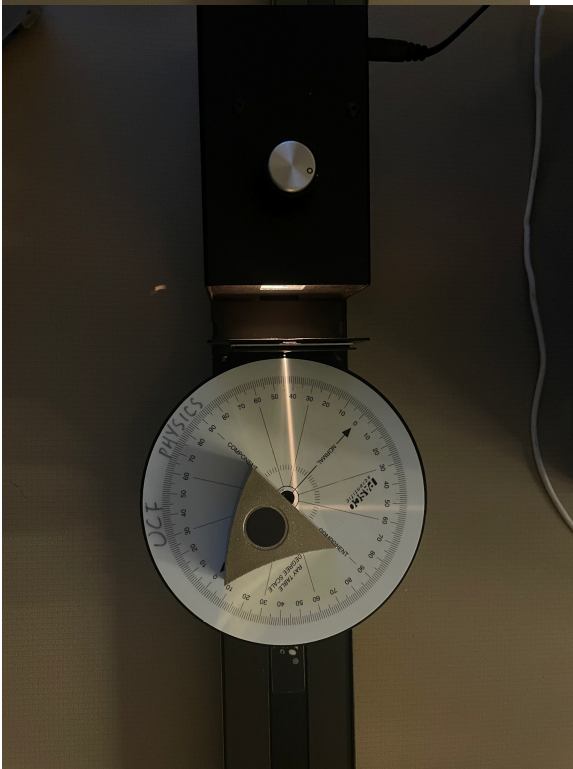
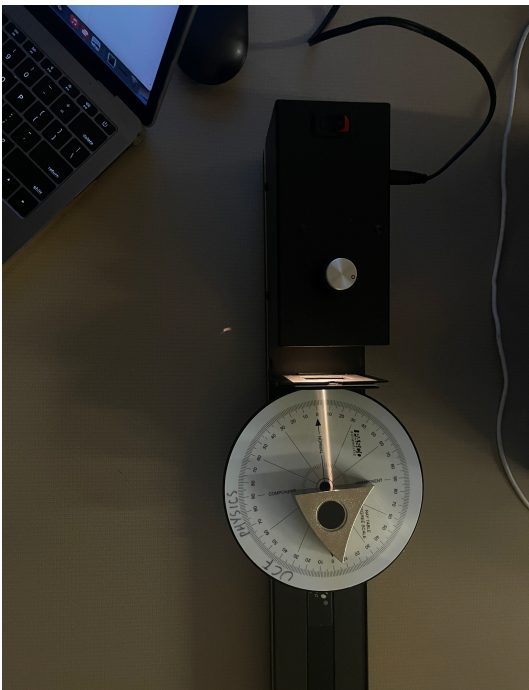
Procedure

Set up the equipment as shown in Figure 1. Adjust the components so a single ray of light is aligned with the bold arrow labeled “Normal” on the Ray Table Degree Scale. Carefully align the flat reflecting surface of the mirror with the bold line labeled “Component” on the Ray Table. With the mirror properly aligned, the bold arrow on the Ray Table is normal (at right angles) to the plane of the reflecting surface. **Include a picture of your set-up in your lab report.**

Rotate the Ray Table and observe the light ray. The angles of incidence and reflection are measured with respect to the normal to the reflecting surface.



Include at least two pictures of your experiments below for Part 1.



By rotating the Ray Table, set the angle of incidence to each of the settings shown in Table 1.

1. For each angle of incidence, record the angle of reflection (Reflection 1).

Table 1

Angle of Incidence (Θ_i)	Angle of reflection (Θ_r)
0°	0
10°	20
20°	40
30°	60
40°	80
50°	100
60°	120
70°	140
80°	160
90°	180

2. Repeat your measurements with the incident ray coming from the opposite side of the normal.
For each angle of incidence, record the angle of reflection (Reflection 2) in Table 2.

Table 2

Angle of Incidence (Θ_i)	Angle of reflection (Θ_r)
0°	0
10°	20
20°	40
30°	60
40°	80
50°	100
60°	120
70°	140
80°	160

90°	180
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3. Are the results for the two trials the same? If not, to what do you attribute the differences?

Yes the angles of the two trials are the same.

4. Part of the law of reflection states that the incident ray, the normal and the reflected ray all lie in the same plane. Discuss how this is shown in your experiment.

The plane the rays lie on are all falling along the surface of the disc, this plane is the 2 dimensional xy plane

5. What relationship holds between the angle of incidence and the angle of reflection?

The angle of incidence is $\frac{1}{2}$ the angle of reflection

6. The drawing shows a top view of an object located to the right of a mirror (Figure 2). A single ray of light is shown leaving the object. After reflection from the mirror, through which location, A, B, C, or D, does the ray pass? Explain your answer. **Hint: Double click the image to be able to draw the rays hitting and bouncing off the mirror**

C, the angle of the ray of incidence is 45 degrees so the angle of reflection is 90 degrees which when followed would intersect with point C

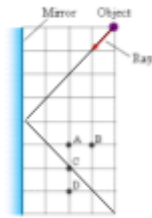


Figure 2

When parallel light rays strike a smooth, plane surface, the reflected rays are parallel to each other. This type of reflection is known as specular reflection. An irregular surface reflects the light rays in various directions. This type of reflection is known as diffuse reflection.

Part 2: Image Formation in a Plane Mirror

Next you will do the following experiment to understand the image formation in a plane mirror.

Equipment:

- Optics Bench with Light Source

- Ray Table and Base
- Component Holder
- Slit Plate
- Ray Optics Mirror

Introduction

The nature of the image you see in a mirror is understandable in terms of the principles you have already learned: the Law of Reflection and the straight-line propagation of light. In this experiment you will investigate how the apparent location of an image reflected from a plane mirror relates to the location of the object, and how this relationship is a direct result of the basic principles you have already studied.

Procedure

Set up the equipment as shown in Figure 3. Adjust the Slit Plate and Light Source positions for sharp, easily visible rays. **Include a picture of your set-up in your lab report.**

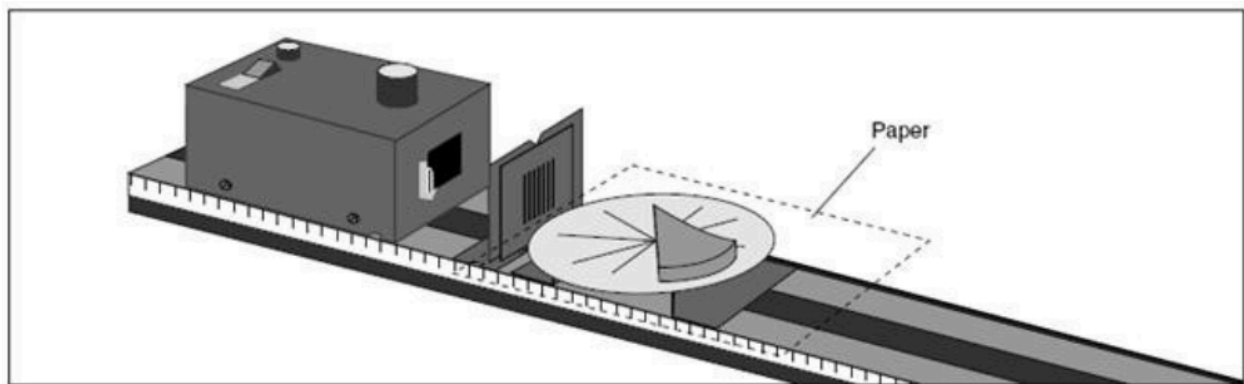


Figure 3

As shown, place a blank, white sheet of paper on top of the Ray Table, and place the Ray Optics Mirror on top of the paper. Position the mirror so that all of the light rays are reflected from its flat surface. Draw a line on the paper to mark the position of the flat surface of the mirror. Look into the mirror along the line of the reflected rays so that you can see the image of the Slit Plate and, through the slits, the filament of the Light Source. (Rotate the mirror as needed to do this.)

1. Are the rays parallel before hitting the mirror?

Before hitting the mirror the rays are parallel. This is because the slit plate and light source produce narrow, straight beams of light that travel in straight lines. Having parallel rays helps clearly show the law of reflection, as each ray reflects off the mirror

at an equal angle to its angle of incidence. This setup allows us to accurately observe and analyze the formation of the virtual image in the plane mirror.

2. Do one of the following:

a. Option 1:

- i. Take a picture from directly above and trace all lines so they match the diagram shown below.

OR

b. Option 2:

- i. With a pencil, mark two points along one edge of each of the incident and reflected rays.
- ii. Label the points (r_1, r_2 , etc.), so you know which points belong to which ray. For example, the top ray will have two dots before hitting the mirror, each which are labeled as r_1 . After bouncing off the mirror, you will label two other points as r_1 for this same ray.
- iii. Repeat this for all rays.
- iv. Remove the paper and reconstruct the rays as shown (Figure 4), using a pencil and straightedge. If you need to, tape on additional pieces of paper.
- v. Draw dotted lines to extend the incident and reflected rays. Include a picture of the lines you drew in this lab report.

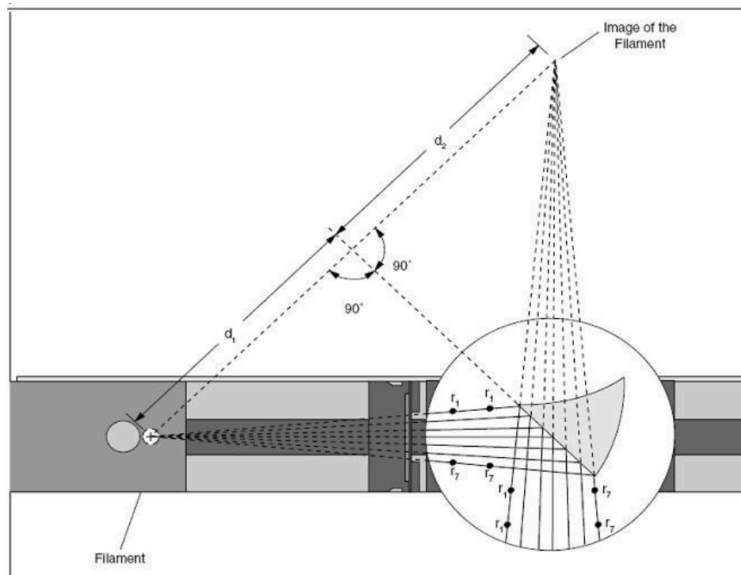
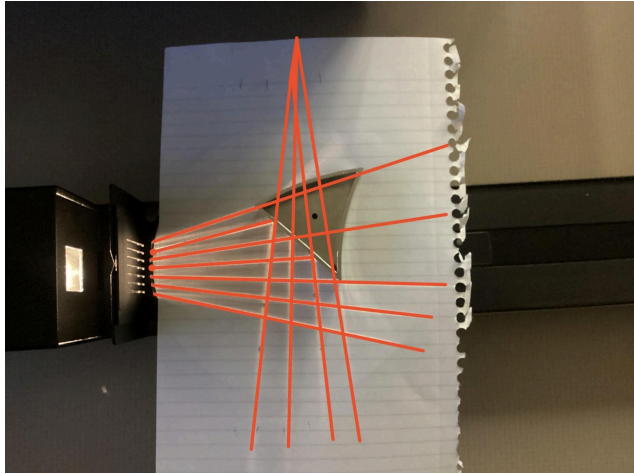


Figure 4



3. How does the perpendicular distance from the filament to the plane of the mirror (distance d_1 , as shown in the Figure 4) compare to distance d_2 ?

In the diagram the filament's light is reflected at a 90° angle by the mirror, forming an image on the screen. Due to this right-angle reflection and the symmetry of the setup, the perpendicular distance from the filament to the mirror (d_1) is equal to the distance from the mirror to the image of the filament (d_2). This equal spacing results from the geometric properties of the reflected rays and the mirror alignment.

4. What is the relationship between object and image location for reflection in a plane mirror?

In a plane mirror the image of an object appears to be the same distance behind the mirror as the object is in front of it. The image is virtual (not formed by actual light rays), upright, and laterally inverted (left and right are reversed).